R code for Data Science for Beginners

Day 2: Individual Exercise

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Task 1. `if...else` function

Mary and John's families (each family has 5 members) are going to a movie and have a hard time deciding who will sit where in a row with 10 seats. You decide to help them sort it out. Please let the Mary's sit in the number 1-5 and the John's in the number 6-10.

Please use `if...else` function to complete the task.

```r
### Task 1-1. Answer
#First, I wanted to create the family lists and a total guest list
MaryFamily=c("Mom","Dad","Jackson","Bella", "Mary")
JohnFamily=c("Mother","Father", "Wayne", "Ophelia","John")
GuestList=c(MaryFamily,JohnFamily)


#Next I am selecting a random guest
Guest=sample(GuestList, 1)
print(paste("The Selected Guest is: ", Guest))
```

```
[1] "The Selected Guest is:  Mary"
```

```r
#Next I want it to make the console say where they should sit
if(Guest %in% MaryFamily) {
  print("The Guest is in Mary's Family and should sit in seat 1-5.")
} else {
  print("The Guest is in John's Family and should sit in seat 6-10.")
}
```

```
[1] "The Guest is in Mary's Family and should sit in seat 1-5."
```

```r
#If you wanted to assign everyone a seat, you could make another table
SeatAssignment= rep(0,10) #I am making a new list

AvailableSeatsForMaryFamily=c(1,2,3,4,5) #I am trying to say which values can be assigned to
AvailableSeatsForJohnFamily=c(6,7,8,9,10)
MaryStart=1
JohnStart=1

for(i in seq_along(GuestList)){
if(GuestList[i] %in% MaryFamily) {
  SeatAssignment[i]= AvailableSeatsForMaryFamily[MaryStart]
  MaryStart=MaryStart+1
} else {
  SeatAssignment[i]=AvailableSeatsForJohnFamily[JohnStart]
  JohnStart=JohnStart+1
}}

MovieSeatingTable= data.frame(SeatAssignment=SeatAssignment,
                              GuestList=GuestList)

MovieSeatingTable
```

```
   SeatAssignment GuestList
1               1       Mom
2               2       Dad
3               3   Jackson
4               4     Bella
5               5      Mary
6               6    Mother
7               7    Father
8               8     Wayne
9               9   Ophelia
10             10      John
```

Task 1-2.

You want two family members to know each other better so decide to mix them up. Please write a script to let one family to sit next to the other family members in a row. Again use the if..else function.

```
### Task 1-2. Answer
SeatAssignment2= rep(0,10) #I am making a new list

AvailableSeatsForMaryFamily2=c(1,3,5,7,9) #To make them intermingle, I assigned evens and odd
AvailableSeatsForJohnFamily2=c(2,4,6,8,10)
MaryStart2=1
JohnStart2=1

for(i in seq_along(GuestList)){
if(GuestList[i] %in% MaryFamily) {
  SeatAssignment2[i]= AvailableSeatsForMaryFamily2[MaryStart2]
  MaryStart2=MaryStart2+1
} else {
  SeatAssignment2[i]=AvailableSeatsForJohnFamily2[JohnStart2]
  JohnStart2=JohnStart2+1
}}

MovieSeatingTable2= data.frame(SeatAssignment2=SeatAssignment2, GuestList=GuestList)

MovieSeatingTable2 = MovieSeatingTable2[order(MovieSeatingTable2$SeatAssignment2), ]

MovieSeatingTable2
```

```
   SeatAssignment2 GuestList
1                1       Mom
6                2    Mother
2                3       Dad
7                4    Father
3                5   Jackson
8                6     Wayne
4                7     Bella
9                8   Ophelia
5                9      Mary
10              10      John
```

Task 2. loop

2-1

We are now in the year of 2022. Please use for loop to print out all the years starting from 2012 to 2022.

```
### Task 2-1. Answer
year=rep(2012:2022)

year
```

```
 [1] 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022
```

2-2

Please use `paste()` function to write a complete sentence like "The year is 2012".

```
### Task 2-2. Answer
yearCallout=sample(year,1) #I wanted it to pick a random year for my sentence

print(paste("The year is", yearCallout, "."))
```

```
[1] "The year is 2013 ."
```

2-3

Turns out we don't really like 2020 and 2021 because Covid messed up many parts of our lives. Please don't print out these two years using the next function.

```
### Task 2-3. Answer
for(nonCovid in 2012:2022){
  if(nonCovid==2020 || nonCovid==2021){
    next
  }
print(nonCovid)
}
```

```
[1] 2012
[1] 2013
[1] 2014
[1] 2015
[1] 2016
[1] 2017
[1] 2018
[1] 2019
[1] 2022
```

Task 3. functions

Please write a function that will always add 10 in addition to whatever number you put in.

```
### Task 3-1. Answer
add10= function(x) {return(x+10)}

add10(23)
```

```
[1] 33
```

```
add10(456)
```

```
[1] 466
```

```
add10(24601)
```

```
[1] 24611
```

Please write a function that will always identify a missing value in your vector. And please write a complete sentence to show where the missing value is located in the vector

```
### Task 3-2. Answer
missingValue= function(x) {print
  (paste
    ("The missing value is located at number",
      which(is.na(x)),
      "in the lineup of the list."))}
```

```r
test=c(1,NA,3,4,5,6,7,8,9)

test2=c("a","b","c","d","e",NA,"g","h","i","j")

missingValue(test)
```

```
[1] "The missing value is located at number 2 in the lineup of the list."
```

```r
missingValue(test2)
```

```
[1] "The missing value is located at number 6 in the lineup of the list."
```

Finally, execute the entire contents of this file. Make sure that you don't get any error message. If you get an error message, it's probably because you forgot to comment out something.